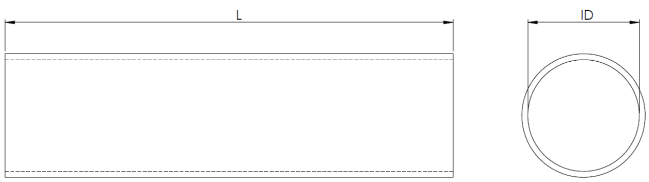
FRP Filament-Wound Pipe

1 – 36 in. SIZE RANGE

150 psi ALL SIZES

5 · 10 · 15 · 20 ft STOCKED LENGTHS

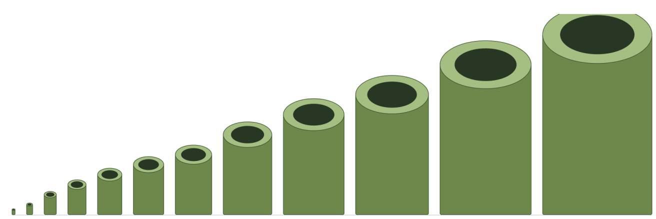
Filament-wound to ASTM D2996 with alternating ±55° helical and 90° hoop windings in the structural layer, designed on a 2:1 hoop-to-axial stress ratio and a minimum 10:1 safety factor. Plain-end as standard.



ID	nominal inside diameter	**P**	pressure rating
T	total wall thickness	**L**	length

LAMINATE COMPOSITION — FILAMENT-WOUND PIPE ONLY — the four right-hand columns are the plies that make up the total wall thickness T. Contact-moulded fittings use the same layer structure to a different schedule; see page 03.

ID nominal dia.	DN metric	P pressure	T wall thk.	T wall thk.	veil C-veil + Nexus	CSM chopped strand mat	struct. roving 2400 tex	ext. C-veil + UV
in	—	psi	in	mm	mm	mm	mm	mm
1	DN25	150	1/4	6.3	0.5	3.5	2.0	0.3
1-1/2	DN38	150	1/4	6.3	0.5	3.5	2.0	0.3
2	DN50	150	1/4	6.3	0.5	3.5	2.0	0.3
3	DN80	150	1/4	6.3	0.5	2.5	3.0	0.3
4	DN100	150	1/4	6.3	0.5	2.0	3.5	0.3
6	DN150	150	1/4	6.3	0.5	2.0	3.5	0.3
8	DN200	150	1/4	6.3	0.5	2.0	3.5	0.3
10	DN250	150	5/16	7.9	0.5	2.0	5.1	0.3
12	DN300	150	3/8	9.5	0.5	2.0	6.7	0.3
14	DN350	150	7/16	11.1	0.5	2.0	8.3	0.3
16	DN400	150	7/16	11.1	0.5	2.0	8.3	0.3
18	DN450	150	1/2	12.7	0.5	2.0	9.9	0.3
20	DN500	150	1/2	12.7	0.5	2.0	9.9	0.3
24	DN600	150	5/8	15.8	0.5	2.0	13.0	0.3
30	DN750	150	3/4	19.0	0.5	2.0	16.2	0.3
36	DN900	150	7/8	22.2	0.5	2.0	19.4	0.3



Nominal inside diameter (in.) — diameters to true relative proportion; stub length illustrative

NOTES

- Standard length 20 ft; also stocked in 5 ft, 10 ft and 15 ft.
- Corrosion-resistant barrier excluded from the strength calculation.
- All wall thicknesses shown include the corrosion barrier.

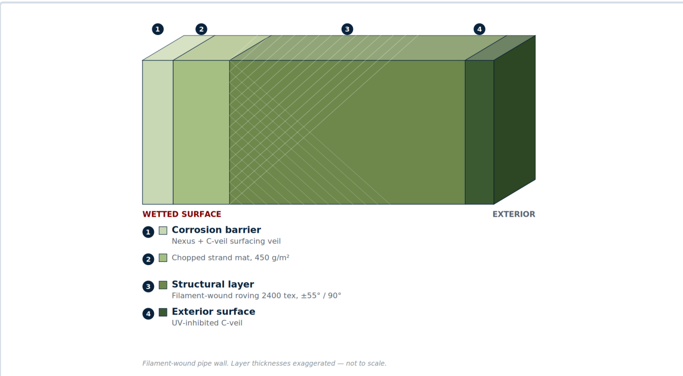
JOINT PREPARATION

- Plain-end as standard.
- Flanged, bell-and-spigot and butt-and-strap preparations on request.
- Stocked in Derakane 411-350 and Derakane 470-300.

Resins, Laminate & Standards

Every component is built as a layered laminate. The layer names below are those used by ASTM C582 and ASME RTP-1 and are the common vocabulary of the industry. What differs between products is not the structure but the ply build-up and the way the structural layer is laid down. Wall thicknesses quoted throughout this catalogue are total thicknesses and include the corrosion barrier.

LAMINATE CONSTRUCTION



Filament-wound pipe wall. Layer thicknesses not to scale.

Inner surface
A resin-rich layer reinforced with a surfacing veil — glass, synthetic or carbon. Nominally 0.010–0.020 in. This is the surface the process fluid touches.

Interior layer
Chopped strand mat, two plies minimum. With the inner surface it forms the **corrosion-resistant barrier** — the part of the wall that does the chemical work, and the part excluded from the strength calculation.

Structural layer
Carries the pressure and the mechanical load. Filament wound on pipe; contact moulded — hand lay-up — on every fitting. The first ply against the barrier is always mat.

Outer surface
A resin coat, paraffinated for a full surface cure. UV-inhibited exterior standard on pipe, available on fittings.

Ply-by-ply build-up differs by product and method. The filament-wound pipe schedule is tabulated on page 07; fitting schedules are held by the manufacturer — see the scope note below.

RESIN SYSTEMS

Derakane 411-350
Bisphenol-A epoxy vinyl ester

HDT	105 °C / 220 °F	TENSILE	86 MPa / 12,000 psi
MODULUS	3,200 MPa	ELONGATION	5–6 %

The standard offering. Broad resistance to acids, alkalis, bleaches and solvents.

Derakane 470-300
Epoxy novolac vinyl ester

HDT	150 °C / 300 °F	TENSILE	85 MPa / 12,500 psi
MODULUS	3,600 MPa	ELONGATION	3–4 %

Specified where elevated temperature, solvent or oxidiser service (chlorine, flue gas) demands higher thermal and chemical resistance.

HDT is a material property, not a service rating. Maximum service temperature is governed by the chemical environment — confirm against the resin manufacturer's chemical resistance guide before specifying.

STANDARDS OF MANUFACTURE AND REFERENCE

CGSB 41-GP-22	Process equipment — reinforced polyester, glass fibre. Governing Canadian fabrication standard; covers tanks, pipe and ductwork, and establishes the 10:1 safety factor.	**ASTM D2996**	Filament-Wound "Fiberglass" Pipe. Governs the pipe on page 07.
ASTM D5421	Contact-Molded "Fiberglass" Flanges. Governs the stub flanges and manway flanges.	**ASTM C582**	Contact-Molded Reinforced Thermosetting Plastic Laminates. Governs the laminate in all contact-moulded fittings.
ASME RTP-1	Referenced for bolt-hole drilling compatibility with RTP-1 vessel nozzle and manway patterns only — see the scope note below.	**ASME B16.5**	Class 150 bolt-hole drilling, NPS ½ through 24.
ASME B16.1 Cl. 125	Bolt-hole drilling, NPS 30 and 36.	**ASME B16.9**	Geometric basis for long-radius elbows (R = 1.5 × ID).
ASME B18.22.1	Type A Narrow plain washers — governs back-face (BF) diameters.	**ASME B1.20.1**	Taper pipe threads — FNPT couplings.

DESIGN BASIS

- Minimum **10:1 safety factor** on the structural laminate, per CGSB 41-GP-22.
- The corrosion-resistant barrier is excluded from the strength calculation — the safety factor applies to the structural wall only.
- Filament-wound structural layers are designed on a **2:1 hoop-to-axial** stress ratio.

GASKETS & BOLTING

- 1/8 in. full-face elastomeric gasket, Shore A60 ± 5, minimum design seating stress (y) 100 psi, gasket factor (m) 1.
- Flat-face mating flanges only — never bolt FRP to a raised-face metallic flange without a full-face adaptor.
- Tighten in a star pattern in equal stages. FRP flanges are damaged by over-torquing — ask us for torque values.

STANDARDS SCOPE

- **ASME RTP-1 is a vessel standard** rated to 15 psig. It does not cover piping or pipe fittings. Where we cite it, we mean bolt-hole drilling compatibility with RTP-1 vessel nozzle and manway patterns — nothing here is offered as an RTP-1 certified vessel component.
- We are a stockist. Item-specific laminate schedules and conformance documentation come from the manufacturer against a specific order — ask at enquiry stage.

FRP DEPOTS

CANADA'S DIRECT SOURCE FOR FRP COMPONENTS

How to order

Send your enquiry to **sales@frpdepots.com** or call **613-499-4950**. Include the following and we will quote from stock where we can.

01 Product type and nominal diameter

02 Pressure rating required

03 Resin system — D411 or D470

04 Quantity and length (pipe, couplings)

05 Any non-standard drilling, stub length or joint preparation

06 Drawings or job specification, if you have them

> Full dimensional data, tolerances and design basis are published in **FRPD-SPEC-001 Revision R3**. Request the current revision with your enquiry.

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